Alfred P. Sloan Jr. (1875-1966)

Alfred P. Sloan, a one-time electrical engineer, became one of the greatest organizational geniuses in American industrial history by helping General Motors recover from its stock crisis. This led to him being promoted to the presidency of General Motors on May 10, 1923. He was then elected board chairman of General Motors in 1937. On April 2, 1956, he announced his resignation after 38 years of dedication to General Motors. Today the Alfred P. Sloan Foundation stands as his legacy.

> Bill Gates (Business@ the speed of thought Warner Books, 1999):
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> I think Alfred Sloan's My Years with General Motors is probably the best book to read if you want to read only one book about business. It's inspiring to see in Sloan's account of his career how positive, rational, information-focused leadership can lead to extraordinary success.

Background

The modern corporation developed during the century spanning 1850-1950. Before 1840, large, coherent organizations were unknown. Business was carried out by companies, partnerships, and trading firms that were limited geographically or used only very loose controls for managing distant operations (e.g., Astor's fur-trading empire). Alfred D. Chandler, the preeminent historian of the growth of the modern firm, said, "A

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businessman of today would find himself at home in the business world of 1910, but the business world of 1840 would be a strange, archaic and arcane place. So, too, the American businessman of 1840 would find the environment of fifteenth-century Italy more familiar than that of his own nation seventy years later."[1] This document is intended to serve as the basis of class discussion and is not intended to illustrate either effective or ineffective management.

The beginnings of large-scale organization can be found in the railroads. In the decade of the 1850s, tens of thousands of miles of rail were laid, creating the basic land transportation system for the eastern portion of the United States. For the first time in recorded history, large volumes of traffic moved more quickly than a horse.

The speed and complexity of railroads created the need for a new kind of administrative system. Scheduling, setting rates, managing the flow of traffic, maintaining the system, and hiring, training, and assigning personnel all required a full-time professional staff that could work with numbers and records and with formal paper-work methods of communication. The concepts of cost-control, return on investment, periodic reporting, line and staff-much of the stuff of modern management-were developed by engineers faced with the problem of running railroads in the 1850s.

In a railroad, there was little argument about who made policy. The central office created the rules for running the railroad, and local managers implemented those rules. The senior management of a railroad was normally divided by function-finance, legal, traffic, and transportation. This functional form of organization was, in turn, applied to other industries such as steel, chemicals, and automobiles.

In 1921-22, there was a sudden, dramatic drop in consumer demand. Few managers of large industrial firms had experience with such a recession-it devastated whole industries. General Motors was forced to drastically write down the value of its inventories and faced financial ruin.

Struggling with the problem of adjusting stocks, production, and assets in an uncertain world, Du Pont and General Motors (both under the direction of Pierre du Pont) developed what later became known as the multidivisional form of organization. At Du Pont (chemicals), the problem was an inflexible, centralized functional structure. By contrast, at General Motors, the problem was forging a company out of a loose alliance of automobile and parts makers.

At General Motors, the chief architect of the new structure was Alfred P. Sloan. Self-described as "specializing in management," Sloan brought an engineer's clarity to industrial administration.

Sloan and Durant

Born in New Haven in 1875, Alfred Sloan's father was a tea, coffee, and cigar merchant. After the age of ten, he grew up in Brooklyn. He earned his B.S. in electrical engineering from M.I.T. in 1895, graduating after only three years of crushing course loads as the youngest member of his class. He went to work at Hyatt Roller Bearing. When Hyatt faced financial failure in 1899, Sloan's father advanced $5000 and let Alfred try to turn the situation around. He did, and wound up in control of a small, healthy, growing firm. Much of the company's health came from the expanding demand for carriages and motorcars. At Hyatt, Sloan came to know a number of fledgling motorcar companies and

the entrepreneurs who founded them. Key among them was William Durant. Durant, a leading wagon manufacturer, had reorganized the failing Buick Motor Company in 1904. In 1908, Buick outsold Ford, building 8487 cars to Ford's 6181. Between 1908 and 1910, Durant merged twenty-five firms with Buick, forming General Motors. The new nameplates that survived were Cadillac, Olds, and Oakland (later Pontiac). The other combinations involved parts and accessory producers. The financing for this combination overextended the company, and Durant was forced to resign in 1910. The next year, Durant and Louis Chevrolet started the Chevrolet Motor Company, which was combined with General Motors several years later.

In the decade that followed, Henry Ford introduced the Model-T and set about reducing unit costs with enormous skill and determination. Quickly surpassing General Motors, Ford be-came the pre-eminent automaker in the U.S. Although General Motors grew in this period of great consumer demand for automobiles, it operated as a loose confederation of firms without a coherent central charter or administration.

In 1916, Durant returned to manage the enterprise again. Still an acquirer, he called Sloan and offered to buy Hyatt. Sloan sold Hyatt, receiving cash and stock in General Motors. Sloan became CEO of United Motors, a firm created to integrate Hyatt and other parts-making firms acquired at that time, including Dayton Engineering Laboratories (DELCO).

> I was of two minds about Mr. Durant. I admired his automotive genius, his imagination, his generous human qualities, and his integrity. His loyalty to the enterprise was absolute... But I thought he was too casual in his ways for an administrator, and he overloaded himself. Important decisions had to wait until he was free, and were often made impulsively. [p. 25] [2]

Sloan's early observations of GM focused on the lack of systematic control over expenditures. He noted that it was a case of "competition among the divisions for available capital and of different preferences at the top." [p. 27] He went on to say that "neither the Executive Committee nor the Finance Committee had the needed information or the needed control over the divisions. The divisions continued to spend lavishly and their requests for additional funds were met." In May 1920, a special executive committee developed a list of maximum permissible expenditures for each division. However, "the division managers failed to stay within their authorized limits on inventory or capital expenditures, and nothing was done effectively to control them. This was decentralization with a vengeance." [p. 30]

In September of 1920, demand for automobiles collapsed. Ford cut prices by 20 – 30 percent. Within a month, General Motors was short of cash. All the car producing divisions shut down, except Buick and Cadillac. Durant resigned. The nation entered recession.

Re-Designing General Motors

With Durant's resignation, Pierre du Pont became president, and Sloan joined the fourperson executive team at the top of the company.

> At the close of 1920 the task before General Motors was reorganization. As things stood, the corporation faced simultaneously an economic slump on the outside and a management crisis on the inside.

> The automobile market had nearly vanished and with it our income. Most of our plants and those of the industry were shut down ... We were loaded with high-priced inventory and commitments at the old inflated price level. We were short of cash. We had a confused product line. There was a lack of control and of any means of control in operations and finance, and a lack of adequate information about anything. In short, there was just about as much crisis, inside and outside, as your could wish for if you liked that sort of thing. [p. 42]

The first order of business was to gain control of General Motors' spending. du Pont and Sloan formed a small team of executives who, in turn, required division executives to submit monthly and quarterly forecasts of production, purchases, and labor. Authority to spend funds was granted only after the forecast (budget) was approved by the central committee. Over time, these forecasts were tied to annual projections, which were expanded to include prices, capital investment, and projected return on investment. All forecasts were developed in coordination with staff estimates of demand and sales, based on national income, seasonal variation, and industry factors. Central staff's forecasts and projections were continually adjusted based on operating data. Sales reports from dealers arrived every two weeks, and the company gathered monthly information on the number of new cars registered in each state. During the 1920s, management in large corporations became increasingly professionalized. Accounting became a college subject, and managers formed professional societies. In this environment, the new planning and control devices developed at General Motors diffused rapidly. In short order, General Electric, Westinghouse, Sears Roebuck, and others adopted similar systems of using statistics, numbers, forecasts, and performance data to manage and control operations.

> The need for financial controls grew out of crises. Controls were brought in to ensure that crises did not recur. Their effectiveness was demonstrated particularly in the depression year 1932. The corporation's U.S. and Canadian unit volume in that year was 50-percent less than that of 1931 and 72-percent below the high of 1929. But the corporation was not demoralized as it had been in 1920 and it stayed in the black. Not many corporations did as well.
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> Financial control as worked out by General Motors gave the corporation a review of operations that reduced the need to administer operations from the top. Central-office management was able to know whether the decentralized management was operating well or poorly and had a factual basis for a judgment regarding the future of any particular part of the business. [p. 148]

The second order of business was product policy, or what today would be called strategy. The immediate competitive problem was Ford, which had over 60% of the automobile and truck market; Chevrolet had 4% and was losing money. Some senior executives at General Motors were pushing for a product to meet Ford head-on, the so-called "revolutionary car." Sloan thought this unwise. "With Ford in almost complete possession of the low-price field, it would have been suicidal to compete with him head on. No conceivable amount of capital short of the United States Treasury could have sustained the losses required to take volume away form him head on." [p.69] Rather than argue against the revolutionary car concept directly, Sloan took an indirect tack: he worked to define the purpose of the company. Exploiting the financial crisis, he got the executive committee to declare that "The primary object of the corporation ... was to make money, not just to make motor cars." [p. 64]

> Every enterprise needs a concept of its industry. There is a logical way of doing business in accordance with the facts and circumstances of an industry, if you can figure it out. If there are different concepts among the enterprises involved, these concepts are likely to express competitive forces in their most vigorous and most decisive form...
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> I wanted to broaden the concept of the product to the concept of the business. I believe it was for this reason that we on the special committee first idealized the problem. We started not with the actual corporation but with a model of a corporation, for which we said we would state policy standards...
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> The product policy we proposed is the one for which General Motors has now long been known. We said first that the corporation should produce a line of cars in each price area ... second, that the price steps should not be such as to leave wide gaps in the line ... and third, that there should be no duplication of the corporation in the price fields or steps...
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> We proposed in general that General Motors should place its cars at the top of each price range and make them of such a quality that they would attract sales from below that price ...
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> In the perspective of so many years gone by, the idea of this policy seems pretty simple ... But it certainly did not seem simple at the time, when Ford had more than half the market with two grades... For all we knew then, our policy might not have worked best. If the industry had thought it would work, the others would have adopted it at the time. The same policy was available to all, but for a number of years General Motors alone was to pursue it and prove its worthiness. [p. 58-69 passim]

The new product policy implied major changes. It called for a car in the low-priced field, priced above Ford's car but of better quality. The policy of "no duplication" also meant a complete rationalization of the then-current product line, as most General Motors divisions offered cars in the same densely populated mid-price range.

The third critical order of business was organization. Sloan's vision of the overall organization was a "happy medium between the extremes of pure centralization and pure decentralization." The new policy asked that the corporation neither remain as it was, a weak form of organization, nor become a rigid, command form.

The organization Sloan designed balanced a number of conflicting forces. The design principles were these::

1. There was a clear line between the operating divisions and the corporate functions. The heads of operating divisions were fully responsible for their operations, but they did not make policy. Corporate executives, without direct operating responsibility, made policy. The point of this design was to keep policy-making from being a purely political struggle for resources.
2. All policies were reviewed by committees.
3. The CEO had ultimate authority but worked, whenever possible, through the committee structure.

4. Divisions were constructed so that meaningful return-on-investment measures could be made. Sloan believed that the key corporate decision was the allocation of resources to businesses and that this could only be accomplished in a rational way if clear measures of performance were available.
5. Although each division had responsibility for its own operations, various operating committees were expected to achieve corporate-wide efficiencies by coordinating the work of the divisions.
6. The number, membership, and scope of the coordinating committees was set by corporate executives.

Sloan clearly recognized that one could not, at the same time, create fully autonomous divisions and demand corporate-wide coordination-the two objectives were in conflict. Nevertheless, he designed this conflict into the structure, recognizing that achieving the right balance in this regard was the single critical issue facing General Motors.

> The inter-divisional relations committees gave a measure of coordination to the functions of purchasing, engineering, sales and the like. The operations committee, including the general managers, appraised the performance of the divisions. The executive committee, with contacts in all directions, made policy. It sat at the head of operations, responsible to the board of directors-indeed it was a committee of the board-but beholden to the finance committee for its larger appropriations. On the operating side the executive committee was supreme. Its chairman was the president and chief executive officer of the corporation; and he had all the authority he needed to carry out established policy. This was the new General Motors scheme of management from which developments down to this day, through much evolution, have been derived. [p. 114]
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> It is not easy to say why one management is successful and another is not. The causes of success and failure are deep and complex, and chance plays a part. Experience has convinced me, however, that for those who are responsible for a business, two important factors are motivation and opportunity. The former is supplied in good part by incentive compensation, the latter by decentralization.
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> But the matter does not end there. Good management rests on a reconciliation of centralization and decentralization, or "decentralization with coordinated control."
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> Each of the conflicting elements brought together in this concept has its unique results in the operation of a businesss. From decentralization we get initiative, responsibility, development of personnel, decisions close to the facts, flexibility-in short, all the qualities necessary for an organization to adapt to new conditions. From coordination, we get efficiencies and economies. It must be apparent that coordinated decentralization is not an easy concept to apply. There is no hard and fast rule for sorting out the various responsibilities and the best way to assign them. The balance between divisional and corporate responsibilities varies according to what is being decided, the circumstances of the time, past experience and the temperaments and skills of the executives involved.
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> The concept of coordinated decentralization evolved gradually at General Motors as we responded to tangible problems of management. [p. 429]

Sloan and Kettering

In 1923, Pierre du Pont retired from active management, and Sloan was appointed president and chief executive officer of General Motors.

Charles Kettering, a year younger than Sloan, also held a B.S. in electrical engineering, earned at Ohio State. At General Motors, Kettering arrived with Sloan when Durant bought his company, DELCO. From 1920 onward, Kettering ran General Motors Research, the first modern industrial research laboratory.

Kettering's great idea was the copper-cooled engine. He believed that by using copper cooling fins, an internal combustion engine could be built that did not need water cooling. By eliminating the radiator, water pump, associated rust, and belts, engines could be made cheaper, more reliable, and easier to maintain. Kettering convinced Pierre du Pont that this "revolutionary" idea could allow General Motors to compete directly with Ford. A good air-cooled car, he argued, would beat Ford at his own game—cheap, reliable transportation for the masses.

The executive committee handed the problem of introducing the air-cooled engine to the Chevrolet and Oakland divisions. Problems soon arose. Research was supposed to do pilot production and Chevrolet the mass production. Division management wanted to know "who was advisor to whom on production: Research to the car division or the car division to Research?" [p. 76]. William Knudsen, who ran Chevrolet, did not like the whole idea and complained "This copper-cooled car isn't any good.... It won't stand the gaff." [4]

Kettering blamed the Chevrolet division for not backing the new idea. The air-cooled engine, he maintained, would be "the greatest thing that has ever been produced in the automobile world." [p. 76] Work on it continued in the Oakland division, which was more optimistic about its prospects. However, within months, the Oakland division reported that the car had failed its field tests. The Executive Committee cancelled the air-cooled project. Kettering was devastated. Sloan wrote him a letter (signed by the entire Executive Committee) which provides an intriguing picture of Sloan as a manager:

> Dear Mr. Kettering: ....
> In order that your mind may be completely relieved as to the position of the undersigned with respect to the air cooled development, we beg to advise you as follows:-
> 1st . We are absolutely confident in your ability to whip all problems in connection with the development of our proposed air cooled cars.
> 2nd . We will continue to have this degree of confidence and faith in you and your ability to accomplish this task until such time as we come to you and frankly state that we have doubts as to the possibility or feasibility of turning the trick and you will be the first one to whom we will come. [p. 78-9]

Despite this letter, Kettering was not mollified. He proposed to resign, arguing that "I am perfectly aware that we can take any proposition and make out of it a 100% success, provided we do not have to overcome an organized resistance within the Corporation. This is impossible unless the Executive Committee can take it upon themselves to force through an Executive order when they know it is going to be of value to the Corporation." [p. 88]

> Sloan wrote back to Kettering, explaining how he saw the role of top management.
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> The great trouble is that there is an apparent lack of confidence in the copper-cooled car ... That, as I see it, is the real problem before us. It is not the merits of the case, and days and weeks spent on it will not alter the situation. What we have got to do is to make our people see the thing as you see it, and with that accomplished, then there will be nothing more to the problem. I do not think that forcing the issue is going to get us anywhere. We have tried that and we have failed. We have got to go at it in a different manner if we are going to succeed. [p. 90-1]

Sloan believed that managers should base their judgments on facts, including the opinions and judgments of other key managers. If the air-cooled engine had merit, that merit should be observable. In a lengthy note to himself on the subject, he wrote:

> I believe that considerable time has been lost in the development of the air-cooled car through lack of appreciation, undoubtedly on the part of all of us, that certain fundamental facts do not, to any extent, subscribe to the contentions of Mr. Kettering-that everybody in General Motors must be sold on the details of his proposed car. I believe that if he had developed a car and demonstrated its performance ... we would have been farther ahead.... Advanced engineering always, like advanced everything else, brings down upon it the discredit of ridicule of minds who cannot see so far. For all this reason, such engineering must be demonstrated in such a way that the facts must be accepted rather than theories. I do not think there would have been any trouble in Oakland or that any changes would have been suggested had Mr. Kettering waited until he had a car that would demonstrate a reasonable satisfactory performance.

Kettering continued to insist that Sloan, now CEO, push through the air-cooled project. Sloan, should, he demanded, fire those who opposed the project. Sloan resisted, and in the process fleshed out his thinking about how one managed a complex firm like General Motors. He wrote:

> The Executive Committee can remove anybody in the in the organization; it can make any changes in personnel it wants to; it can adopt any policies that in its judgment are desirable for the benefit of the Corporation, but there are certain things it cannot do and that is, it cannot order a co-operative spirit to be developed in the organization-it has got to be done in an entirely different way.

To deal with this situation, Sloan created the General Technical Committee-a high-level advisory body consisting of Kettering, the engineering head of each division, corporate staff engineers, and selected corporate officers. The pursuit of the air-cooled engine slowed after Kettering's laboratory discovered and patented the solution to engine knock: tetraethyl-lead.

> General Motors is an engineering organization. Our operation is to cut metal and in so doing to add value to it. About 19,000 engineers and scientists work in the corporation, of whom 17,000 are in the divisions and 2,000 in the general technical staffs. Many of our leading executives, myself included, have an engineering background. It is natural, therefore, that we should always have understood that our progress is linked to technological progress. The permanent drive of research and engineering in industry is to accelerate technological

progress, to incorporate in products and in manufacturing the advances made in science and technology, and to shorten the time between development and production. To achieve these goals, we long ago differentiated a staff function from the operational function. [p. 248]

Drucker on Sloan

Peter Drucker, "The Best Book on Management Ever," Fortune, 23/4/1990

What, then, are the main lessons in My Years with General Motors, at least as I read Alfred Sloan's intentions? This is how I would paraphrase them:

The first is that management is a profession and that the manager is-or should be-a professional. This may sound trite in 1990-it was far from obvious thirty years ago. But frankly, while most managers by now preach it, not too many yet practice it.

Like any professional, a physician, for instance, or a lawyer, the professional manager has a "client:" the enterprise. He is bound to subordinate his own interests to those of the client. It is duty to the client that characterizes the "professional."

Professionals do not make decisions by opinions nor according to their preferences. They make them according to the facts. ...

The job of a professional manager is not to like people. It is not to change people. It is to put their strengths to work. And whether one approves of people or of the way they do their work, their performance is the only thing that counts, and indeed the only thing that the professional manager is permitted to pay attention to. I once said to Sloan that I had rarely seen more different people than the two men who during my study had run the most profitable divisions of GM, Chevrolet and Cadillac.

"You are quite mistaken", he said. "These two men were very much alike - both performed." But performance is more than the "bottom line." It is also setting an example. And this requires integrity. Limited only to these twin boundaries, business performance and performance as example and mentor, there should be absolute tolerance and indeed the greatest diversity.

Dissent, even conflict, are necessary, are indeed desirable. Without dissent and conflict there is no understanding. And without understanding, there are only wrong decisions. To me the most fascinating parts of Sloan's book are the memoranda in which he first elicits dissent and then synthesizes dissenting views into an understanding, and, in the end, into consensus and commitment.

Leadership is not "charisma." It is not "Public Relations." It is not showmanship. It is performance, consistent behavior, trustworthiness.

Finally-and perhaps the most important lesson-the professional manager is a servant. Rank does not confer privilege. It does not give power. It imposes responsibility.

> One can argue with Sloan's postulates-indeed Sloan very much wanted the reader to argue with them: "Otherwise they won't take them seriously", he once said when I raised the question. But they are why Alfred Sloan wrote the book and why My Years with General Motors is "must" reading.

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[1] Chandler, Alfred D., The Visible Hand, Cambridge: Belknap Press, 1977, p. 455.
[2] Unless otherwise attributed, the boxed quotes are from Alfred P. Sloan, Jr. *My Years with General Motors*, Doubleday, 1964.